

Tribhuvan University

Faculty of Humanities and Social Sciences

A Project Report

On

FITLY EXERCISE AND MEAL PLANNING APP

Submitted To

Department of Computer Application

Xavier International College

Kathmandu

In partial fulfillment of the requirements for the degree of Bachelor of Computer Application

Submitted By

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Under the Supervision of

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that the project entitled "**FITLY EXERCISE AND MEAL PLANNING APP**", prepared under my supervision by Anju Shrestha in partial fulfillment of the requirements for the degree of Bachelor of Computer Application, be accepted for the final evaluation.

……

SIGNATURE

Roshan Jha

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LETTER OF APPROVAL

This is to certify that the project entitled "**FITLY EXERCISE AND MEAL PLANNING APP**", prepared by Anju Shrestha in partial fulfillment of the requirements for the degree of Bachelor of Computer Application, has been evaluated. In our opinion it is satisfactory in the scope and quality of a project for the required degree.

Signature of Supervisor Roshan Jha Faculty Member Department of Computer Application Xavier International College, Kathmandu	**Signature of Coordinator** Rajesh Nepal Coordinator Department of Computer Application Xavier International College, Kathmandu
Signature of Internal Examiner	**Signature of External Examiner**

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our supervisor, Mr. Roshan Jha, for his continuous guidance, valuable suggestions, and constructive feedback throughout the development of this project. His support and encouragement played a significant role in its successful completion.

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Finally, we would like to thank all the faculty members and friends who directly or indirectly supported us during this project.

Yours sincerely,
Anju Shrestha

ABSTRACT

In the contemporary world with its fast pace of life, health and fitness have become standard concerns due to sedentary lifestyles, poor diets, and physical inactivity, all of which have contributed to the rising incidence of obesity, diabetes, and other chronic illnesses. While digital health solutions have become more accessible, most still lack comprehensive, personalized, and holistic platforms that integrate exercise, nutrition, and progress tracking.

This project presents AI-based fitness and meal planning web application designed to help individuals of all fitness levels adopt healthier lifestyles. Built using the MERN (Mongo DB, Express.js, React.js, Node.js) stack, fitly provides scalability, responsiveness, and a modern user experience. Users can input their personal health objectives—such as weight loss, muscle gain, or maintenance— and the system intelligently generates meal and workout plans using machine learning algorithms trained on real-world health and fitness data. The application features a user-friendly dashboard where individuals can track their meals, workouts, and progress. It also provides real-time, dynamic recommendations that adapt to user behavior, making the plans more effective over time. The AI integration ensures continuous personalization, helping users stay motivated and aligned with their goals.

By offering a smart, accessible, and cost-free alternative to personal trainers and nutritionists, *fitly* bridges the gap between expert fitness advice and practical daily implementation. Its mission is to empower users with the tools and insights needed to maintain long-term healthy lifestyles through technology-driven, personalized health guidance.

Keywords: *MERN Stack, Google Gemini API, AI, Personalized Meal Planning, Workout Automation, Full-Stack Web Development, Digital Health & Fitness, Progress Tracking.*

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
BCA	Bachelor of Computer Application
DMR	Dasal Metabolic Rate
ERD	Entity-Relationship Diagram
JWT	JSON Wave Token
LLM	Large Language Model
MERN	Mongo DB, Express.js, React.js. Node.js
ML	Machine Learning
MVC	Model View Controller
PWA	Progressive Web App
REST	Representational Estate Transfer
TU	Tribhuvan University
UI/UX	User Interface/User Experience

CHAPTER 1: INTRODUCTION

1.1. Introduction

In the modern age, with lifestyles dominated by sedentary living and nutritionally unhealthy diet, the importance of physical fitness and healthy diet cannot be stressed more. Rising lifestyle conditions such as obesity, hypertension, diabetes, and cardiovascular diseases highlight the importance of healthy living. However, with individuals not being able to maintain healthy lifestyles due to a lack of time, no one-on-one guidance, and lack of proper knowledge regarding diet and exercise, the majority of them are not able to do so.

Technological innovation has witnessed mobile and web applications being built in an attempt to advance users' health and well-being. Diet planners, fitness trackers, and exercise apps are today's trends. Although trending, most of the platforms either possess generic exercises or distinct services for diet and exercise, creating a fractured user experience.

Fill this gap with a comprehensive solution that combines personalized exercise planning, smart diet advice, and progress monitoring. It relies on AI to adjust plans based on user feedback and activity, making it simpler to develop consistent and efficient routines. With mobile and web usage ongoing in Pakistan and worldwide, fitly wants to be a convenient, easy-to-use health advisor for users in search of sustainable fitness objectives.

1.2. Problem Statement

In the contemporary fast life, people tend to overlook their health due to time, information, or economic limitations. There are fitness and nutrition applications, but the majority of them either offer generic, non-tailor able plans, Lack intelligent feedback or personalization over time or just focus on something—one area—either food or exercises, not both. Such fractured experience dissuades long-term usage, and most abandon it after the first few days because of irrelevance or inability to muster motivation.

Fitly aims to solve this problem by offering an adaptive, one-stop-shop fitness and nutrition solution. AI-powered personalization and tracking enable users to get customized plans based on their individual body measurements, preferences, and goals and with continuous adaptation according to behavior and performance.

1.3. Objectives

The main objectives of this project are:

- Providing AI-designed personalized meal plans that customizes exercise programs as per the physical fitness and schedule availability of the user.
- Encouraging long-term engagement through motivational influences and goal monitoring.

1.4. Scope and Limitations

The scope of the Fitly application is to develop a complete full-stack fitness and meal planning web application built with the MERN stack (Mongo DB, Express, React, Node.js). The application can be run from any contemporary web browser and is desktop and mobile-compatible with responsive layout.

The major scope of the project includes:

- Secure user account and login administration.
- Establishing goals and developing personal habits.
- Artificial intelligence-based meal and workout recommendations.
- Monitoring progress via data visualization dashboards and insights.
- Storage of the daily log, proposed plans, and record of progress.
- Storage of the daily log, proposed plans, and record of progress.

There are some limitations in this system. They are:

- Real-time synchronization with wearable fitness trackers.
- Native mobile apps (e.g., iOS or Android).
- Advanced medical consultations or diagnosis.

1.5. Development Methodology

The development process of the project is modular-based and agile. The application is developed in phases, and there is ongoing integration and testing. The MERN stack was selected due to its efficiency, flexibility, and speed in development.

The Key methodologies are:

- Agile development: Weekly sprints with defined deliverables and continuous evaluation.
- Component-based architecture: Especially on the client side with reusable React components.
- Model-View-Controller (MVC): Applied in backend architecture in order to keep the logic clean.
- Data-driven decisions: AI models are trained on Kaggle datasets with focus areas being fitness metrics and nutritional facts.
- Responsive design guidelines: Making it usable on any device.

The tools and technologies used:

- Front-end: React.js, Tailwind CSS
- Backend: Node.js with Express.js
- Database: Mongo DB Atlas
- AI/ML: Google Gemini AI API for Exercise & Meal planning recommdations.
- Version Control: GitHub
- Deployment: Render (front-end), Render (back-end).

1.6. Report Organization

The report is organized in the following chapters:

- **Introduction:** The first chapter gives the basic introduction of the project. It also includes the problem statement, objectives, scope and limitations, and development methodology.
- **Background Study and Literature Review:** The second chapter describes the background study and the review of existing systems.
- **System Analysis and Design:** The third chapter introduces system analysis and system design. System analysis includes requirement analysis, feasibility analysis, and different modelling diagrams.

- **Implementation and Testing:** The fourth chapter includes the tools used to build the project, and the testing methods and details of the system.
- **Conclusion and Future Recommendations:** The fifth chapter gives the conclusion of the project and suggestions for future improvement.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1. Background Study

Recent advancements in artificial intelligence (AI) and large language models (LLMs) have transformed the capabilities of health and fitness applications. AI-driven personalization, natural language understanding, and recommendation systems have significantly improved the relevance and effectiveness of health advice offered to users. Language models like Google Gemini (formerly Bard), GPT, and other generative AI tools are now being integrated into modern web and mobile applications to deliver context-aware, dynamic content and recommendations.

Studies show that AI-assisted health systems outperform traditional static systems by offering personalized suggestions, adapting to user feedback, and generating human-like insights. These systems can analyze user input in real-time, account for history, preferences, and behavioral trends, and generate meal or workout plans accordingly.

Use of Google Gemini API in Fitly

In Fitly, the Google Gemini API has been integrated to generate dynamic, personalized exercise and meal plan recommendations. Gemini's natural language capabilities allow the system to interpret user profiles, preferences, fitness goals, and even dietary restrictions to create tailored plans. Unlike rule-based or hardcoded recommendation engines, Gemini offers more human-like guidance, supports context switching, and delivers justifications and explanations that improve user understanding and trust. This integration allows Fitly to act not just as a tool, but as a conversational fitness companion. It can answer questions, suggest alternatives, and adapt plans based on evolving goals, making the app more interactive, intelligent, and user-friendly.

2.2. Literature Review

While many fitness and diet platforms provide useful features, they often fall short in offering a truly adaptive, intelligent, and conversational experience. Static plans, lack of flexibility, and minimal feedback mechanisms limit user engagement and long-term effectiveness.

Fitly’s use of the Google Gemini API sets it apart by:

- Enabling dynamic meal and workout plan generation using real-time user inputs.
- Supporting natural conversations for user questions and feedback.
- Enhancing personalization through a powerful LLM-based backend.

This makes Fitly a pioneering application in the domain of AI-enhanced health and wellness platforms.

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CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1. System Analysis

The System analysis was carried out to understand the requirements, users, processes, and overall structure of the Fitly AI platform before implementation. The analysis identified how users interacted with the mobile application to set personalized health goals, generate adaptive exercise routines, calculate daily dietary requirements, and track daily meal intake using AI-driven features. It also identified the administrative requirements for managing users, workout templates, food databases, AI prompt configurations, and overall system data. Based on these requirements, the system functions and interactions were organized for the design and implementation of the platform.

3.1.1. Requirement Analysis

Requirement analysis was performed to identify the functional and non-functional requirements of Fitly AI. The requirements were derived from the main purpose of the platform, which was to provide AI-generated workout and meal planning tailored to an individual's body metrics, dietary preferences, and fitness targets. The requirements were divided into user-side and administrative functions so that the mobile application and web-based admin panel could be designed according to their respective responsibilities. The following requirements were identified to be included in the application.

i. Functional Requirements

The functional requirements described the operations that the system was required to perform. Fitly AI mainly involved two actors: User and Admin. The User interacted with the mobile application to input personal health metrics, receive customized exercise routines and meal plans, log workouts and nutrition, and monitor fitness progress over time. The admin used the web-based administration panel to manage users, update food and exercise repositories, monitor system performance, and maintain AI model integration parameters. The major functions performed by each actor were as follows:

User Registration and Profile Setup

- Email and password can be used to sign up.
- Following authentication, users fill out a simple profile that comprises:
    - Name

- Age
- Gender
- Weight
- Height
- Overall fitness objective (weight reduction, muscle gain, maintenance).

Note: This profile data is not used directly to auto-generate plans. The system collects more detailed data on the specific plan pages.

Meal Plan Input and Generation

On the Meal Plan page, the user must manually input some diet information. It only creates plans after filling in this form.

Required user input:

- Reported weight
- Dietary preferences (e.g., keto, high-protein, vegetarian)
- Food allergies or dietary restrictions (e.g., gluten, dairy, nuts)
- Ideal number of meals per day (e.g., 3, 4, 5)

After submission:

- The system forwards the input to the Google Gemini API
- Gemini returns a personalized meal plan with options, nutrition facts, and scheduling.
- The plan is displayed to the user for viewing or regeneration.

Exercise Plan Input and Generation

On the Exercise Plan page, user input for workout planning is required.

Required input:

- Weight and height
- Existing activity level (novice, intermediate, advanced)

- Preferred exercise days of the week.
- Optional: Focus area (e.g., strength training, cardio)

Once submitted:

- The system sends this information to Gemini
- Gemini returns a customized weekly exercise routine
- The plan includes exercise names, durations, muscle groups, and rest periods

Daily Tracking and Dashboard

- Users can monitor daily workouts and food intake.
- The dashboard displays:
    - Total calories burned and consumed
    - Weekly workout frequency
    - Progress toward goals (e.g., weight loss)

The dashboard updates automatically based on user logs.

Conversational AI Support

- The app includes a Google Gemini chatbot providing:
    - Q&A about exercise or nutrition
    - Alternative exercises or meal suggestions
    - Habit reinforcement and motivational tips
    - AI-generated explanations of plans

This feature enhances interactivity and offers personalized assistance.

ii. Non-Functional Requirements

Non-functional requirements are often called “quality attributes” of a system. Non-functional requirements specify the quality attributes, constraints, and operational criteria that the system must satisfy. These requirements ensure that Fitly AI operates efficiently,

securely, and reliably across both the mobile application and the administrative panel. The primary non-functional requirements are categorized as follow:

- **Compatibility:**
    - Fully compatible with Chrome, Firefox, Safari, and Edge.
    - Responsive design for mobile, tablet, and desktop.
- **Performance:**
    - Plan generation should not exceed 5 seconds.
    - Page loading should not exceed 3 seconds on stable internet.
- **Scalability:**
    - Backend and database should support horizontal scaling.
    - Auto scaling enabled on Render and Mongo DB Atlas.
- **Usability:**
    - Clean, minimalist design with contextual help (tooltips, placeholders)
    - Step-by-step guided inputs for meal and exercise forms
    - Simple dashboards using numbers and charts for clarity
- **Security:**
    - Secure password storage using bcrypt.
    - JWT-based session authentication.
    - API keys stored in environment variables (.env)

3.1.2. Feasibility Analysis

Feasibility analysis is used to find the strengths and weaknesses of a proposed project and to show directions that will improve the project and achieve the desired results. The feasibility analysis of this project was carried out. It is as follows:

i. Technical Feasibility

The system is technically feasible to implement. The tools used (Python, Google Gemini, Mongo DB Atlas, React and MERN Stack) are reliable and freely available, and only an

ordinary computer with a Wi-Fi connection is needed. The system can also be extended to other platforms like mobile and web later, and the technical manpower is easily available.

ii. Operational Feasibility

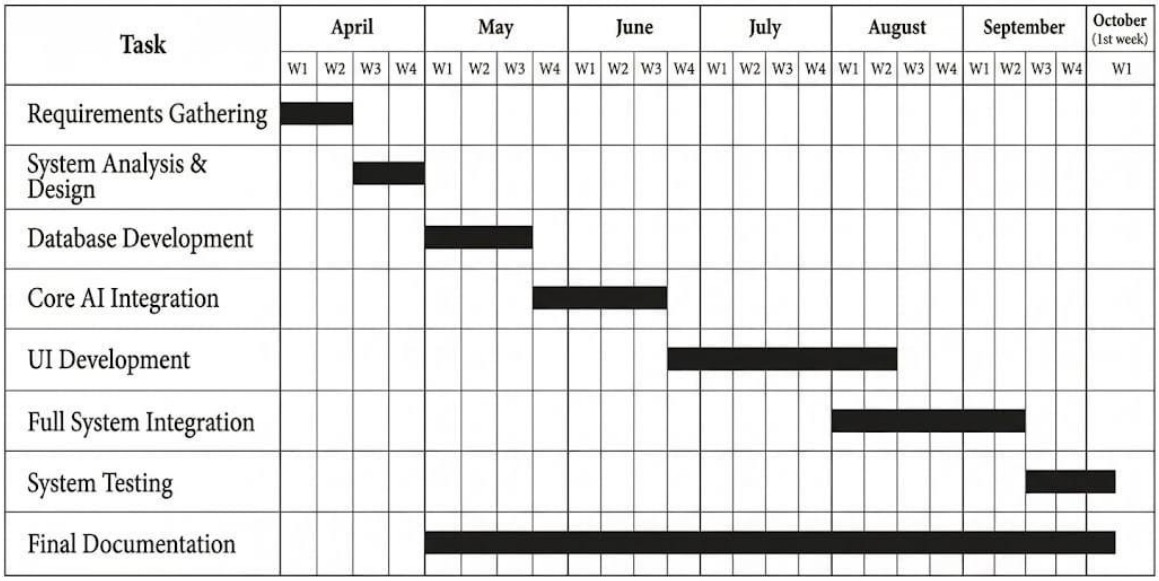
Operational feasibility means whether a system will be used effectively after it has been developed. It examines how well the system solves the user's problem and whether it can be smoothly adopted by both end-users and administrators. The mobile application features an intuitive, user-centric design that simplifies complex fitness metrics, macro calculations, and routine planning into accessible daily interactions. It fulfills a real need of maintaining good body and health, so it is operationally feasible.

iii. Economic Feasibility

Economic feasibility is used to find the financial resources needed for the project. It evaluates the cost-effectiveness and financial viability of developing and deploying the system. The project maintains a low initial capital requirement by leveraging open-source development tools, free-tier cloud databases, and cost-effective hosting platforms. The development of the system is within the budget, and this was possible because most of the technologies used are freely available.

iv. Schedule Feasibility

Schedule feasibility ensures that the project can be completed within a realistic timeframe aligned with academic or deployment milestones. The core functions were done first, followed by testing and improvement. With proper task distribution, the project can be completed within the given academic time. The project timeline is shown in the Gantt chart (Figure 1).

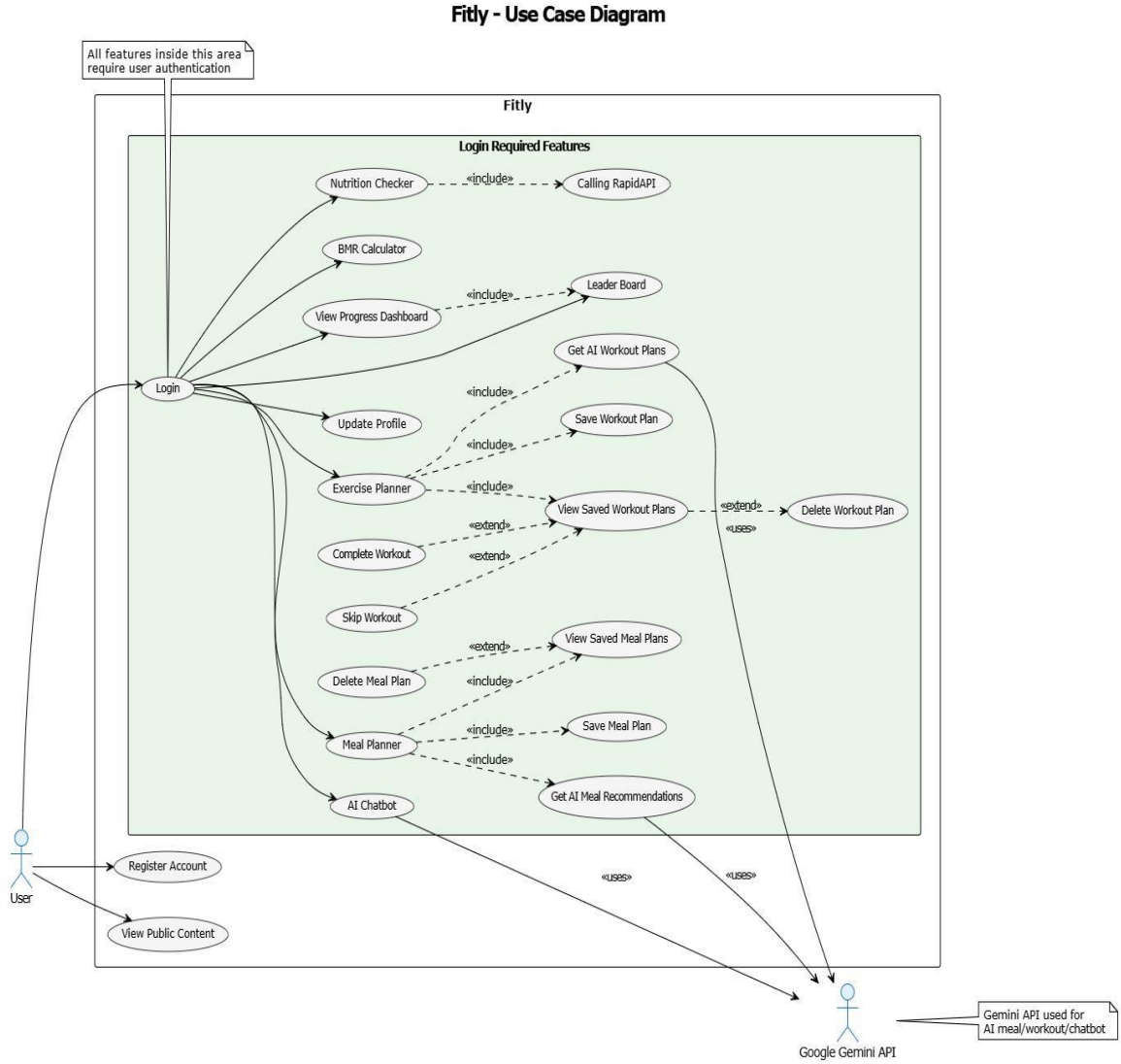


3.1.2: Gantt Chart of FITLY

3.1.2. Software Specification

Component	**Technology/Tool**
Frontend	React.js, Tailwind CSS
Backend	Node.js, Express.js
Database	Mongo DB Atlas
AI Integration	Google Gemini API
Authentication	JWT
Version Control	Git, GitHub
Deployment	Render (Frontend), Render (Backend)
Development tools	Postman, VS Code, and Chart.js

3.1.3 Use Case Diagram



3.1.3: Use case diagram

3.1.4 Use Case 01: Module Selection

Table 3.1.4: Use case diagram for module selection	
Module Selection	
Use case id	UC-01
Purpose	Select the module of your choice

Actors	User
Pre-condition	Application must be launched and installed and user is logged in
Post condition	Selected module screen is displayed

Basic Flow			
	Actor Action		**System Response**
01	Presses BMR calculator	**02**	Display the layout of BMR calculator
02	Enters inputs personal data	**04**	Displays output

Table 3.1.5 Use Case 02: Meal and Workout Generation

Table 3.1.5: Use case diagram for meal and workout generation	
Meal and Workout Generation	
Use case id	UC-02
Purpose	Generate personalized meal plans and workout routines
	using AI
Actors	User, Google Gemini API

Pre-condition	User must be logged in and have completed profile setup
Post condition	Meal plan and workout plan are generated and saved to user's profile

Basic Flow			
	Actor Action		**System Response**
01	User selects meal planner or exercise planner	**02**	Display input form for user preferences
03	User inputs dietary preferences/fitness goals	**04**	Validate input and send request to Gemini API
05	Gemini API processes request	**06**	Generate and display personalized plan
07	User saves the generated plan	**08**	Store plan in database with user association

3.1.6 Use Case 03: Progress Tracking

Table 3.4.3: Use case diagram for progress tracking

Progress Tracking	
Use case id	UC-03
Purpose	Track user's fitness progress and display analytics
Actors	User
Pré condition	User must be logged in and have workout plans created
Post condition	Progress data is updated and displayed on dashboard

Basic Flow			
	Actor Action		**System Response**
01	User selects Dashboard	**02**	Display progress tracking interface
03	User inputs workout completion data	**04**	Store progress data
05	User views progress analytics	**06**	Generate statistics from stored data
07	User checks leaderboard standings	**08**	Display leaderboard with user rankings

3.2. System Design

During the development, we used the following diagrams to understand the requirements of the system.

3.2.1. Architectural Design

System design is the process of defining the elements of a system such as the architecture, modules, components, the interfaces between them, and the data that goes through the system. System design is meant to satisfy the needs and requirements of the system through a well-organized design. Fitly follows 3- Tier Architecture:

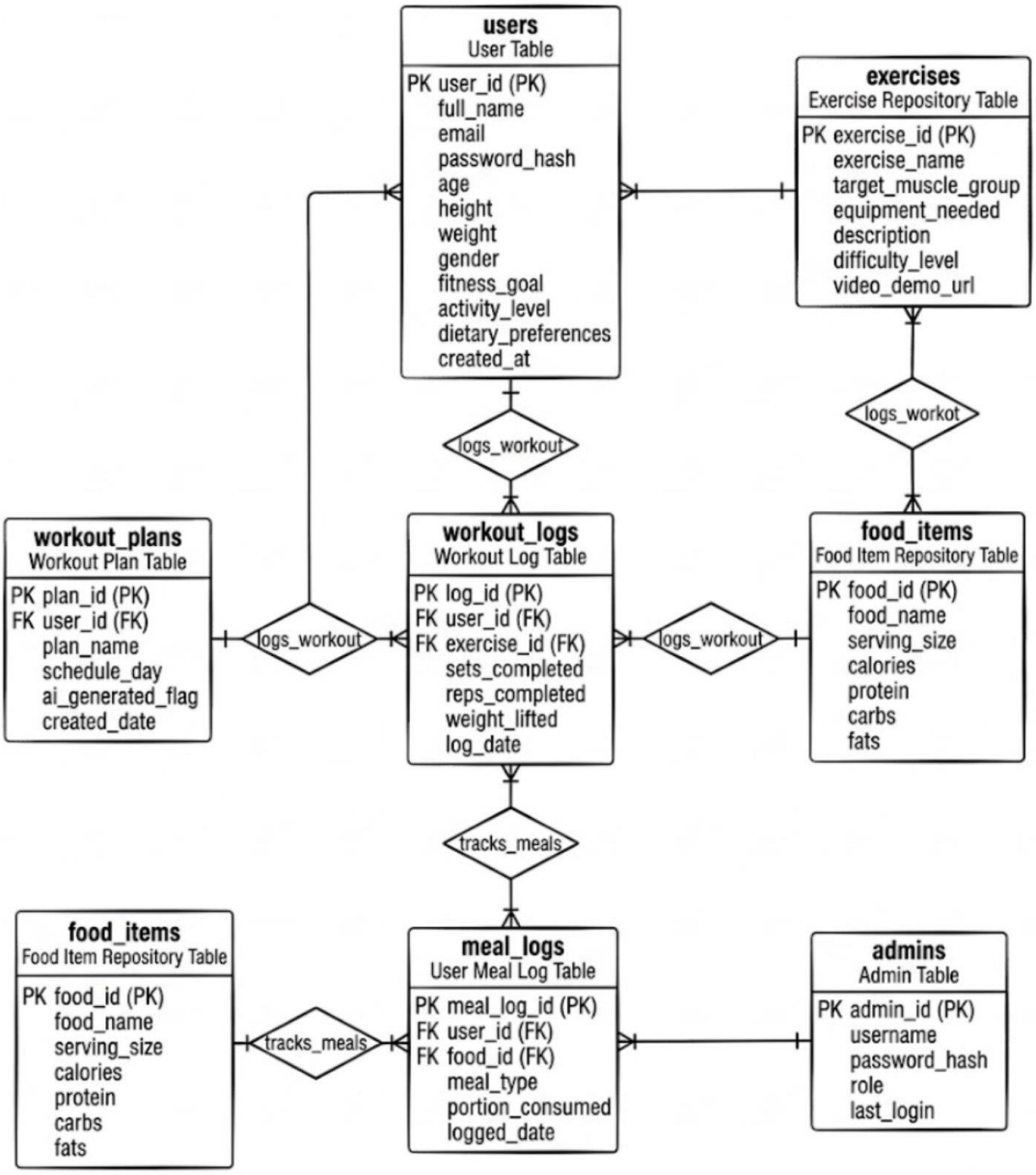
- Presentation Layer (Frontend): Mobile App (Flutter/react native) or Web Client (React/next.js) for user interaction.
- Application/Business Logic Layer (Backend): REST-ful API built with node.js/fastAPI handling authentication, plan generation logic and AI API orchestrations.
- Data Layer: PostgreSQL/Mongo DB for persistence and external AI Service (e.g. Gemini API) for dynamic generation.

3.2.3. Database Design

The database design for Fitly AI outlines how data is structured, stored, and managed to support the mobile application and administrative panel. A relational database management system (RDBMS) approach was selected to ensure data integrity, establish secure

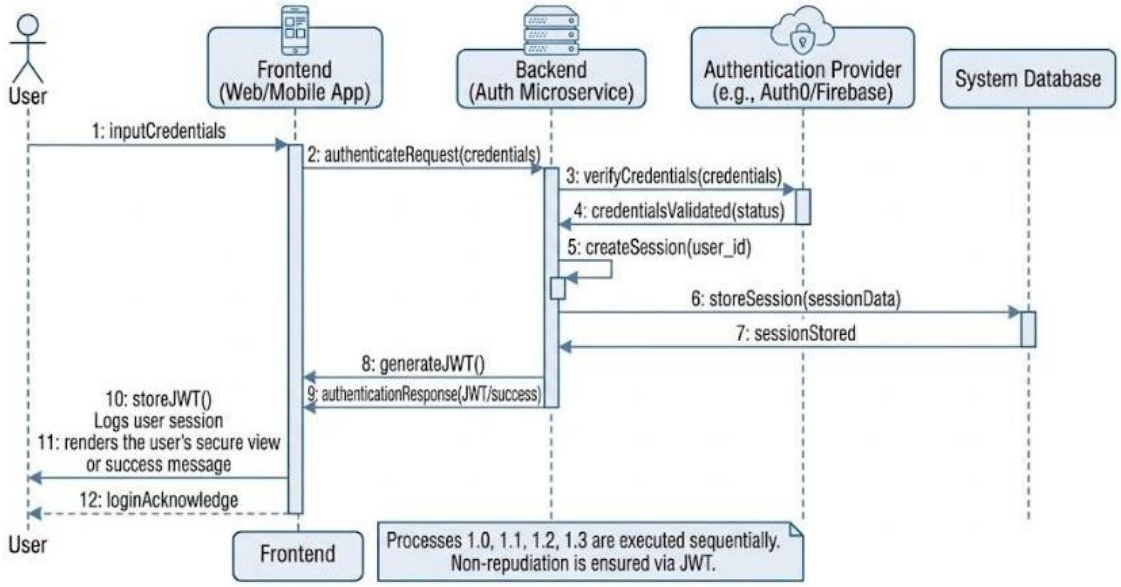
relationships between users and their health metrics, and maintain fast query performance for workout and meal logs.

The database architecture comprises several core entities and tables designed to handle user profiles, AI-generated plans, food repositories, and activity tracking.

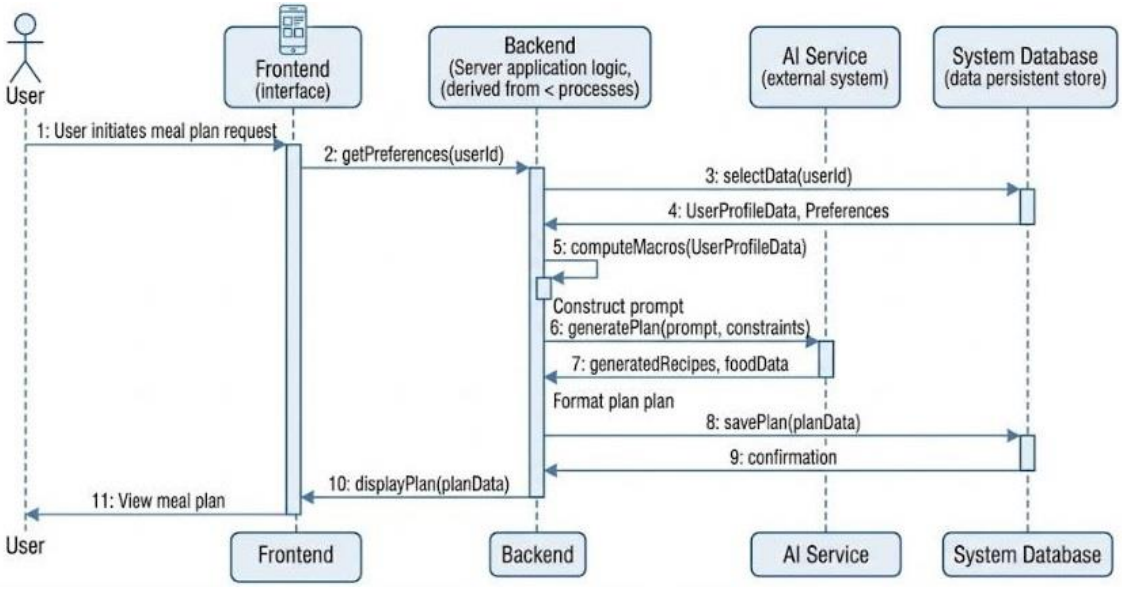


3.2.3: ER diagram of fitly

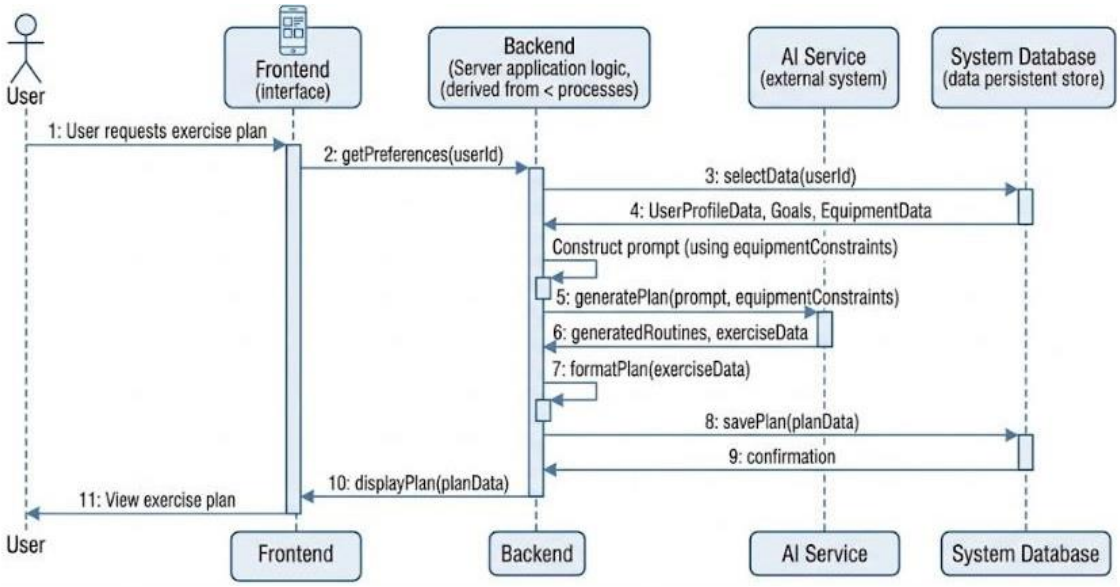
3.2.4 UML Object-Oriented Diagram



3.2.4 User authentication (UC-01)



3.2.4 Meal planner (UC-02)



3.2.4 Exercise planner (UC-03)

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1. Implementation

This chapter explains the procedure adopted to deploy the fitly app. It covers the technologies used across system layers, frontend and backend architecture, the database schema, and how the Google Gemini API was integrated to offer intelligent, context-aware suggestions for fitness and diet plans.

4.1.1. Web Application Platform

Fitly is developed as a full web application using the MERN stack:

- **MongoDB**: Cloud-hosted document-based database.
- **Express.js**: Backend web server and RESTful API handler
- **React.js**: Component-based, dynamic frontend framework
- **Node.js**: Runtime for server-side JavaScript execution

This combination enables modular design, high performance, and efficient development of a single-page application.

4.1.2 Frontend Implementation

The frontend is built with React.js and styled using Tailwind CSS, enabling responsive and modern UI development. It manages user interactions, navigation, and data collection for meal and workout planning.

Key Frontend Features:

- **User Authentication Pages:** Includes Sign Up and Login components with JWT token handling.
- **Meal Plan Input Form:** Users enter data such as weight, dietary preferences, allergies, and meal frequency.
- **Exercise Plan Input Form:** Users input height, weight, activity level, and preferred workout days.
- **Dashboard:** Displays key stats like calories burned, food consumed, and fitness progress using charts and progress bars.

- **AI Chat Assistant Interface:** Enables users to ask questions related to fitness and diet, receiving real-time responses via Gemini integration.

4.1.3 Backend Implementation

The backend is developed using Node.js and Express.js. It manages routing, input validation, user sessions, and communication with external services such as the Gemini API.

Key Backend Features:

- **Authentication APIs:**
    - POST /api/auth/register: Registers a new user with hashed password storage
    - POST /api/auth/login: Authenticates user and returns JWT token
- **Plan Generation Routes:**
    - POST /api/plan/meal: Accepts diet form data and returns a Gemini-generated meal plan.
    - POST /api/plan/exercise: Accepts workout form data and returns a Gemini-generated workout plan.
- **Progress Logging APIs:**
    - POST /api/log/meal: Saves food intake logs.
    - POST /api/log/workout: Saves workout progress logs.

4.1.4 Database Structure

Fitly uses Mongo DB Atlas for storing application data. The schema is designed to be flexible and extendable.

Key Collections:

- **User:** Stores user credentials, profile information, and fitness goals.
- **Plans:** Stores both meal and workout plans generated by the Gemini API, along with the user's original input.

- **Logs:** Tracks user behavior such as meals eaten, exercises completed, and progress over time.

Future enhancements like bookmarks, favorites, and personal notes can easily be integrated into the schema.

4.1.5 Integration of AI with Google Gemini API

A core feature of Fitly is its integration with the Google Gemini API, which enables intelligent, personalized plan creation.

How Gemini Is Utilized:

- **Meal Plan Page:** User inputs (weight, allergies, dietary preferences, number of meals) are formatted into a prompt and sent to Gemini. The API responds with a structured plan containing meal suggestions, nutritional info, and schedule.
- **Exercise Plan Page:** User inputs (height, weight, fitness level, training preferences) are sent to Gemini. The response includes a workout plan with exercises, durations, target muscles, and rest intervals.
- **Chat Assistant:** Users can type natural questions like:
    - "What should I eat after a workout?"
    - "Can tofu replace chicken in a high-protein diet?"
    - "Suggest a 5-day beginner plan."

Gemini interprets the input and provides intelligent responses, turning Fitly into a virtual fitness coach.

4.1.6 Deployment and Hosting

Fitly is deployed using scalable and cloud-native platforms:

- **Frontend**: Hosted on Render, with continuous deployment via GitHub.
- **Backend**: Deployed on Render, with auto scaling and secure handling of environment variables.
- **Database**: Managed through Mongo DB Atlas, offering automatic backups and real-time monitoring.

4.2. Testing

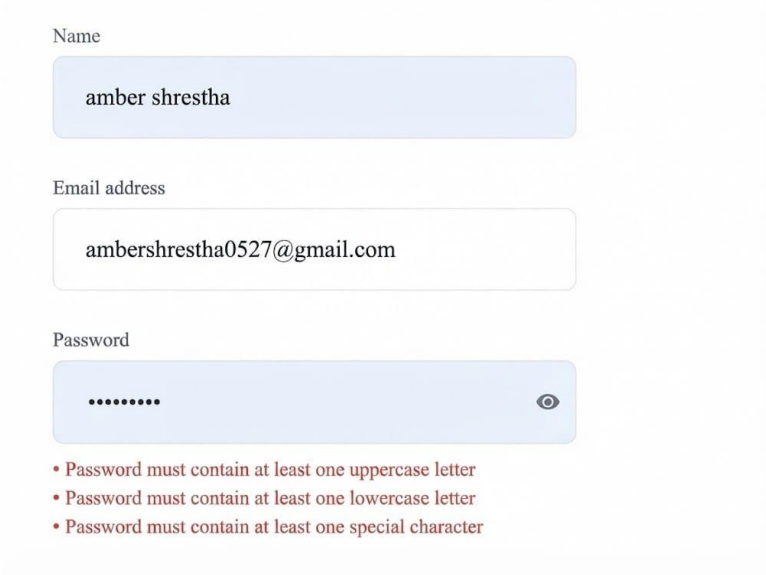
4.2.1 Home Page Load Test

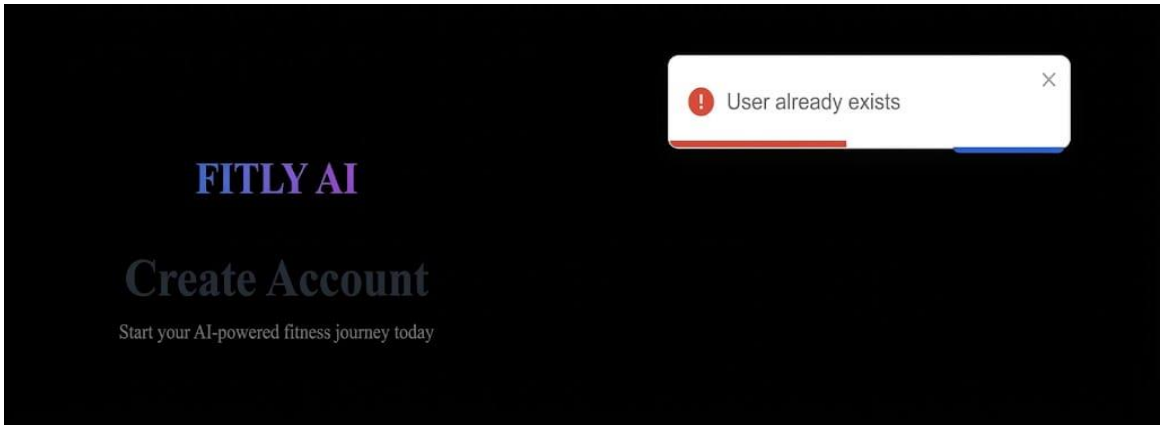
Test ID	Description	Pre-condition	Expected Result	Actual Result
HPL - 001	Initial page load speed	User opens application URL	Page loads within 3 seconds	Page loaded in 2.1 seconds – PASS
HPL - 002	Navigation menu display	Home page is loaded	All menu items are visible and clickable	Navigation menu displays correctly – PASS
HPL - 003	Mobile responsiveness	User accesses on mobile device	Layout adjusts properly to screen size	Mobile layout works perfectly – PASS
HPL - 004	Logo and branding	Home page loads successfully	Logo and branding elements display correctly	All branding elements visible – PASS

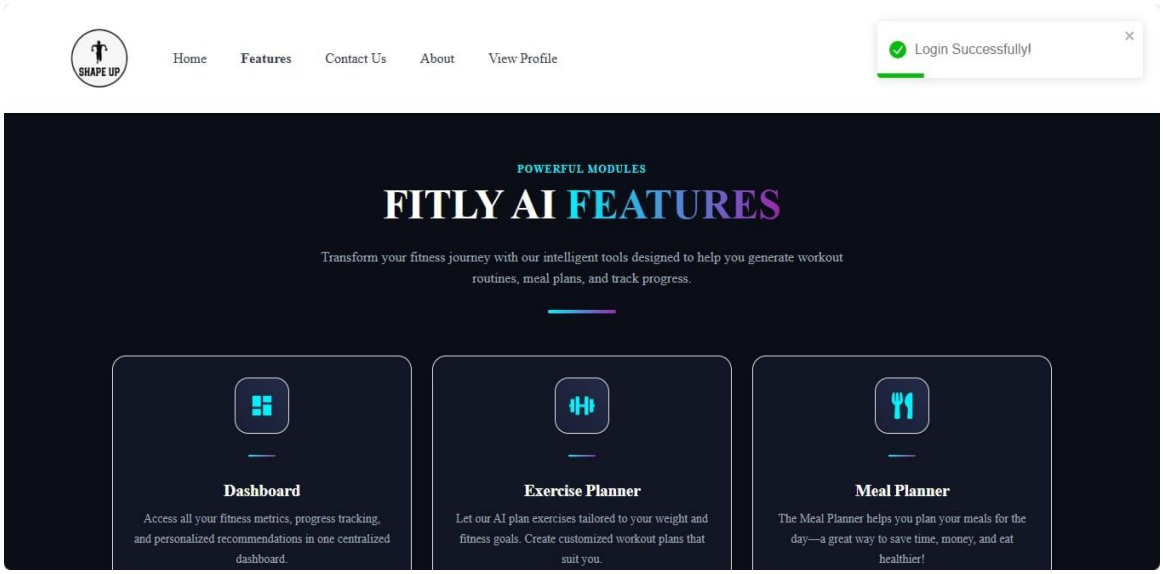
4.2.2 User Registration and Authentication Testing

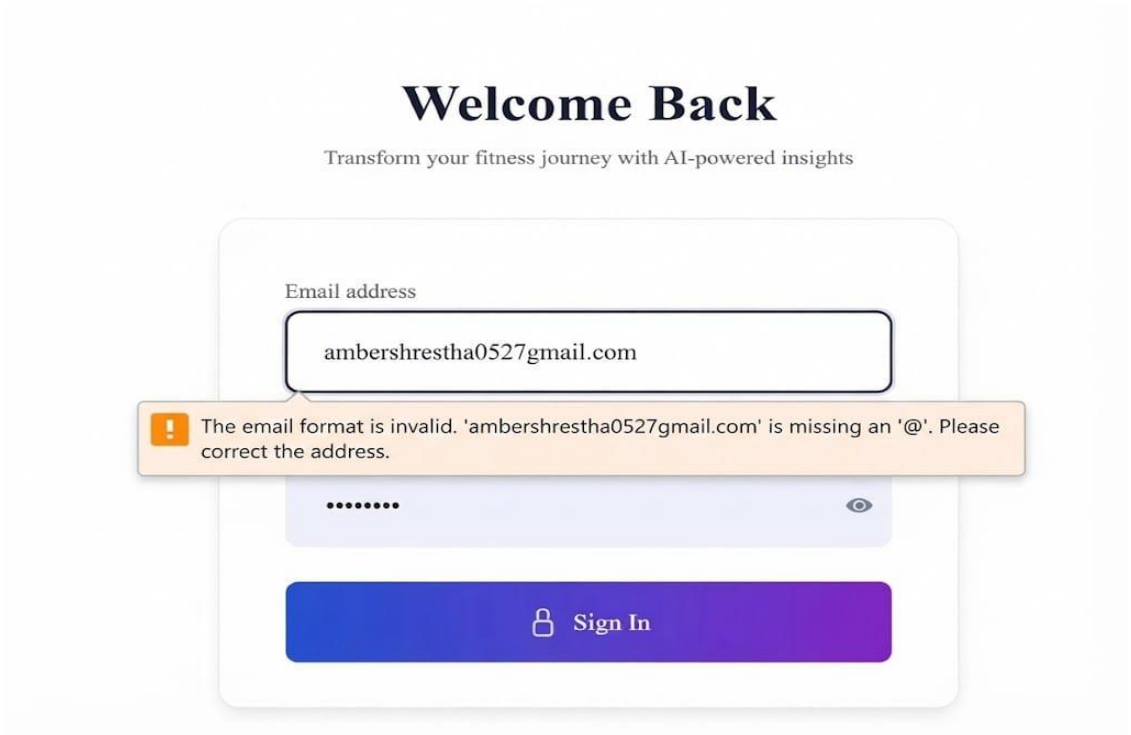
Test ID	Description	Pre-condition	Expected Result	Actual Result
URA-001	Valid user registration	User on registration page	Account created successfully	Registration completed successfully – PASS
URA-002	Duplicate email registration	User enters existing email	Error message: "User already exists"	Proper error message displayed – PASS

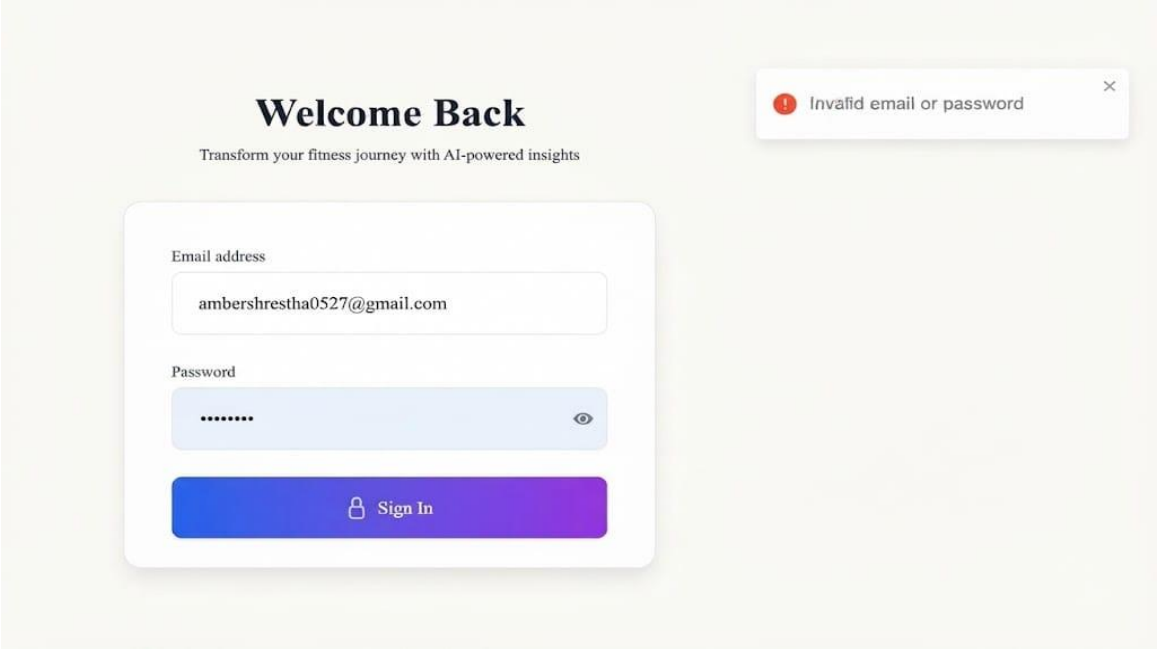
URA-003	Invalid email format	User enters invalid email	Error message for invalid format	Email validation working correctly – PASS
URA-004	Password strength validation	User enters weak password	Password requirements shown	Password strength checker functional – PASS
URA-005	Successful login	Valid credentials entered	User redirected to dashboard	Login successful, dashboard loaded – PASS
URA-006	Invalid login credentials	Incorrect credentials entered	Error message displayed	Authentication error handled properly – PASS




User already exists
FITLY AI
Create Account
Start your AI-powered fitness journey today


SHAPE UP
Home
Features
Contact Us
About
View Profile
Login Successfully!
POWERFUL MODULES
FITLY AI FEATURES
Transform your fitness journey with our intelligent tools designed to help you generate workout routines, meal plans, and track progress.
Dashboard
Access all your fitness metrics, progress tracking, and personalized recommendations in one centralized dashboard.
Exercise Planner
Let our AI plan exercises tailored to your weight and fitness goals. Create customized workout plans that suit you.
Meal Planner
The Meal Planner helps you plan your meals for the day—a great way to save time, money, and eat healthier!

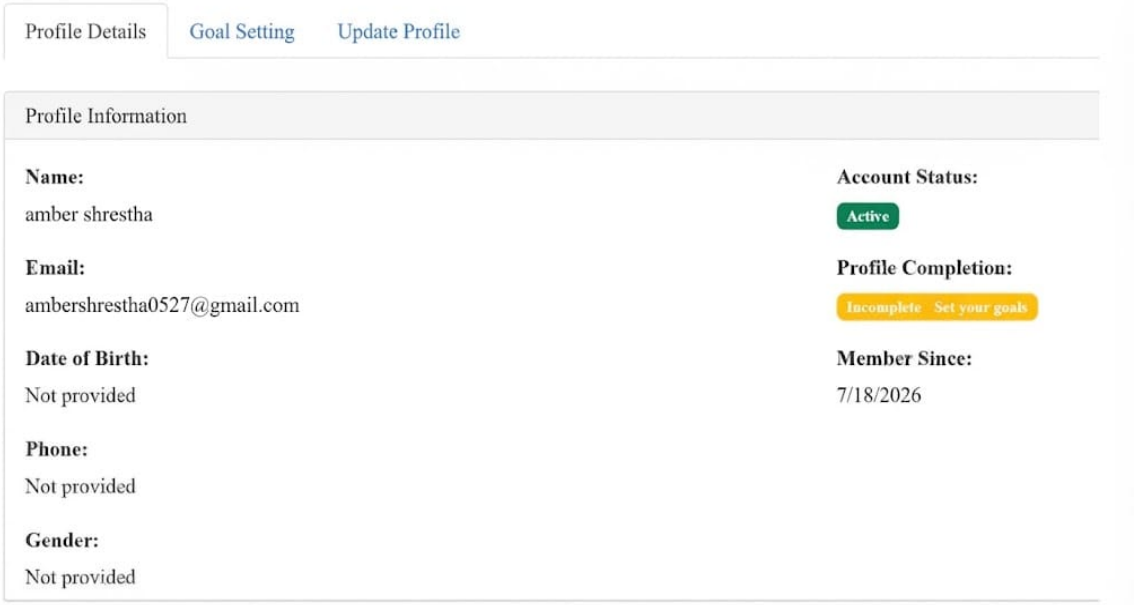

Welcome Back
Transform your fitness journey with AI-powered insights
Email address
ambershrestha0527gmail.com
The email format is invalid. 'ambershrestha0527gmail.com' is missing an '@'. Please correct the address.
Sign In

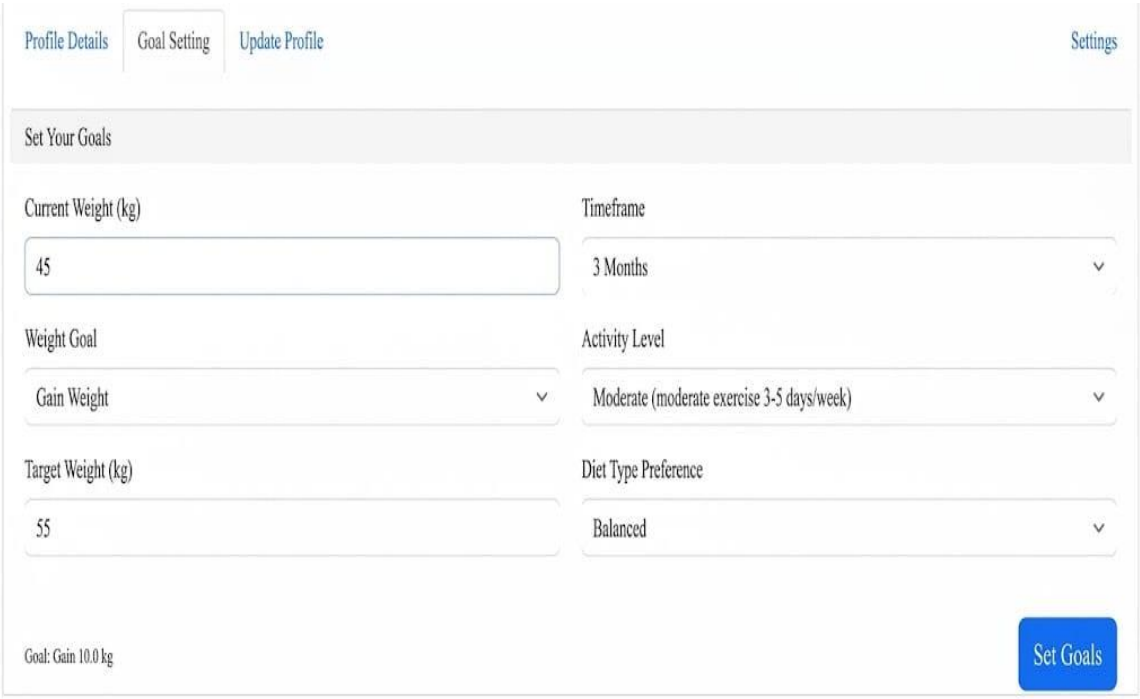


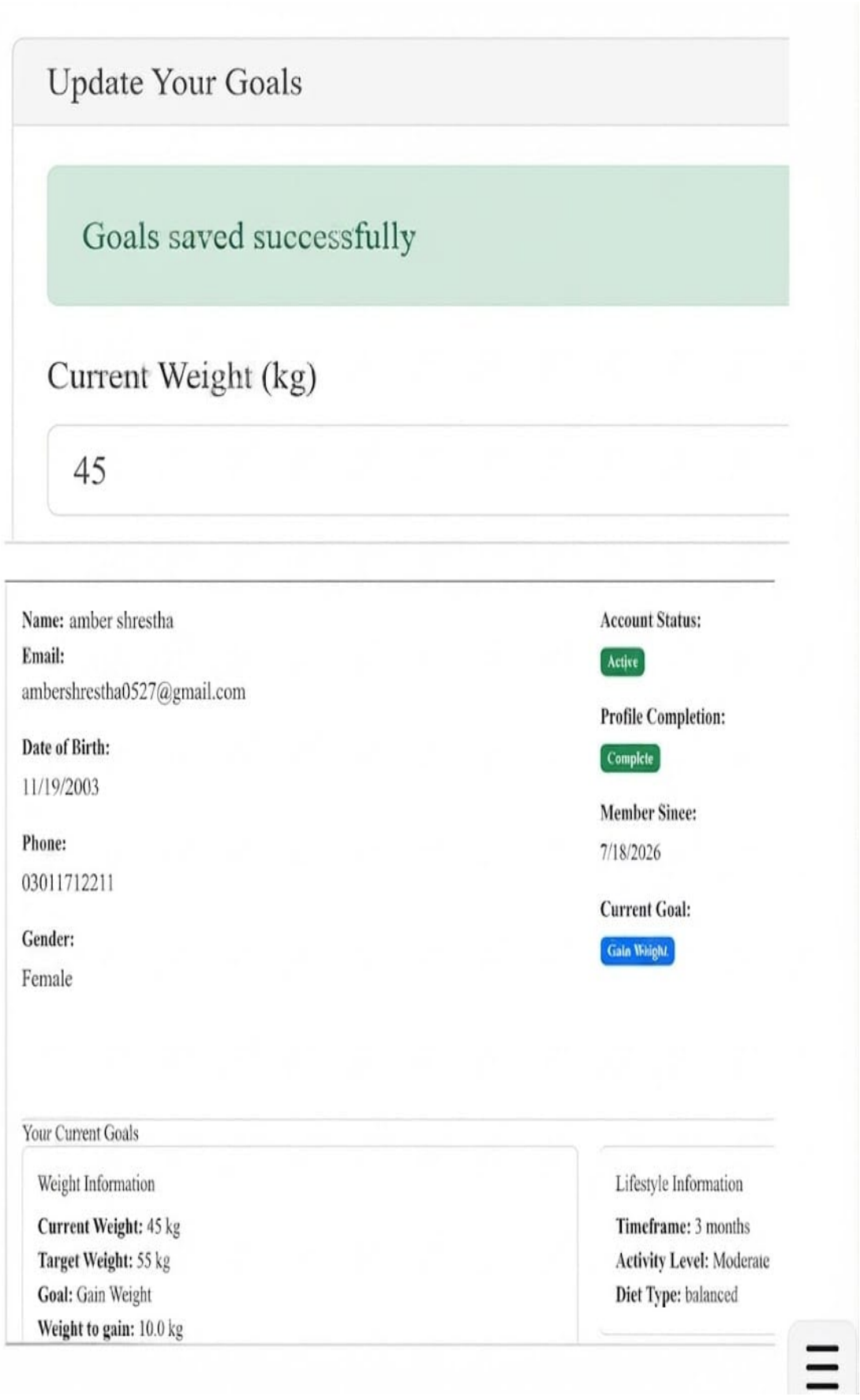
4.2.3 Profile Setup and Goal Assignment Testing

Test ID	Description	Pre-condition	Expected Result	Actual Result
PSG - 001	Profile information entry	User logged in	Personal information saved successfully	Profile data stored correctly – PASS
PSG - 002	Fitness goal selection	Profile setup page loaded	Goals assigned and saved	Goal selection working properly – PASS
PSG - 003	Activity level selection	Profile form accessible	Activity level set correctly	Activity levels saved successfully - PASS

PSG - 004	Dietary preferences setup	Profile setup in progress	Preferences saved properly	Dietary settings configured - PASS

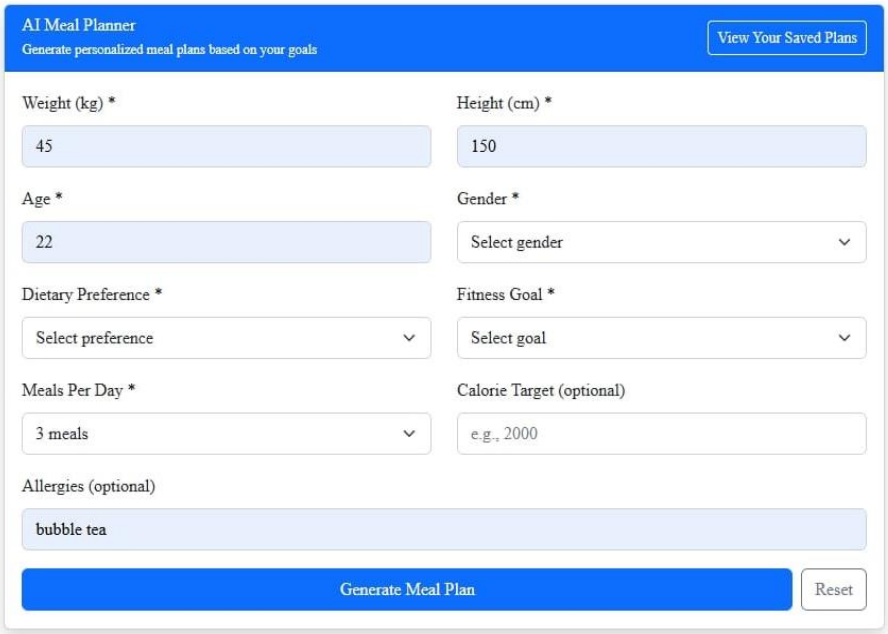


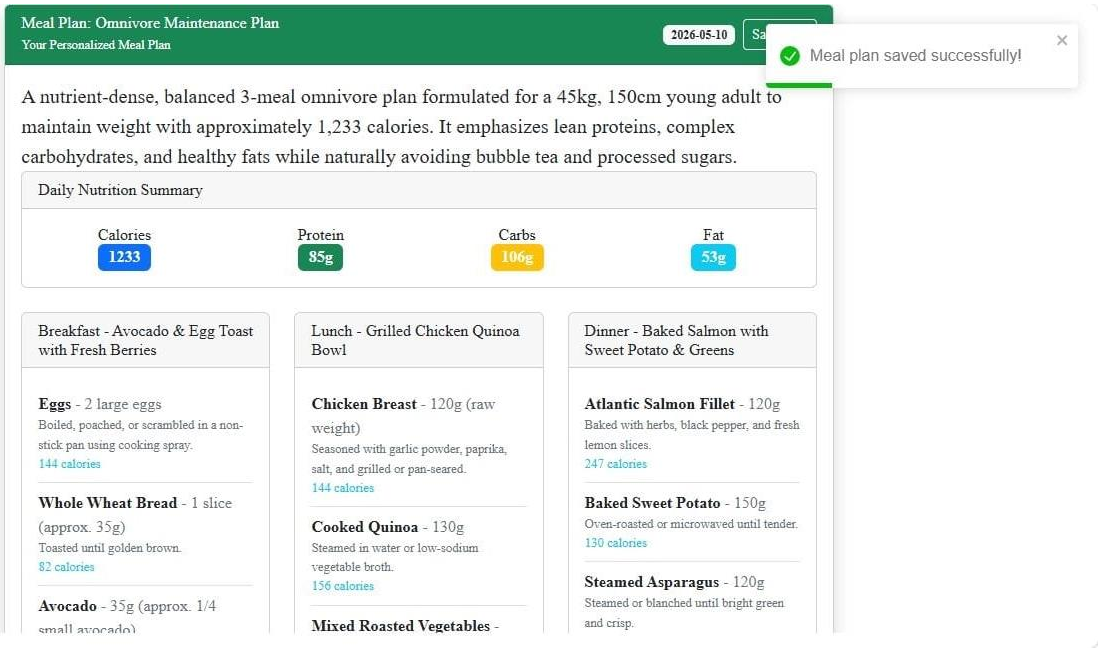



Update Your Goals
Goals saved successfully
Current Weight (kg)
45
Name: amber shrestha
Email:
ambershrestha0527@gmail.com
Date of Birth:
11/19/2003
Phone:
03011712211
Gender:
Female
Account Status:
Active
Profile Completion:
Complete
Member Since:
7/18/2026
Current Goal:
Gain Weight
Your Current Goals
Weight Information
Current Weight: 45 kg
Target Weight: 55 kg
Goal: Gain Weight
Weight to gain: 10.0 kg
Lifestyle Information
Timeframe: 3 months
Activity Level: Moderate
Diet Type: balanced

4.2.4 Meal Plan Generation Accuracy Test

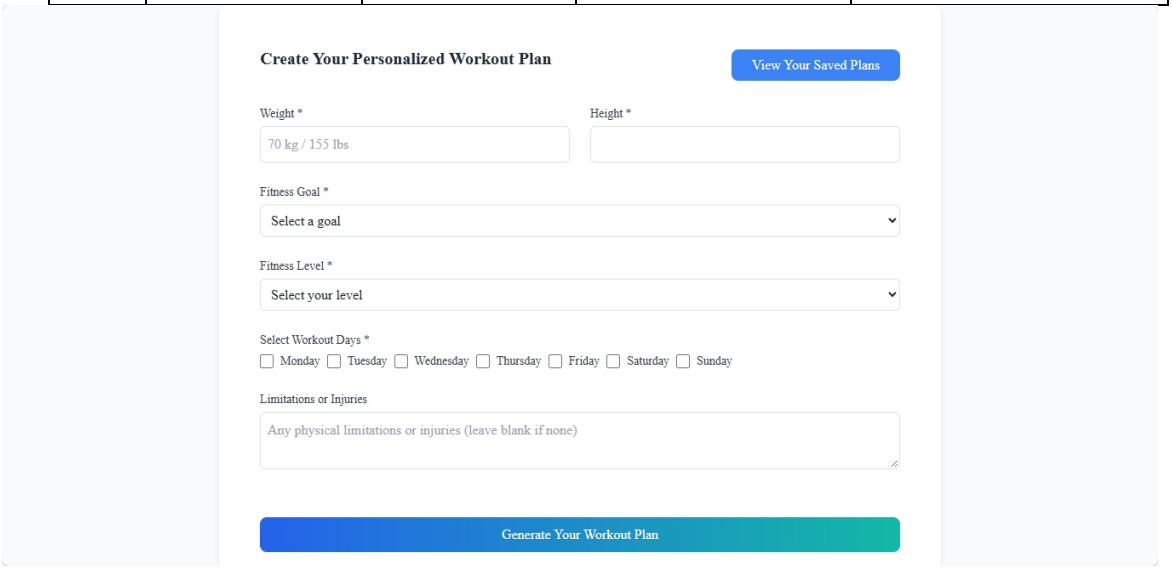
Test ID	Description	Pre-condition	Expected Result	Actual Result
MP G-001	AI meal plan generation	User profile complete	Personalized meal plan generated	Gemini API generated appropriate meal plan – PASS
MP G-002	Dietary restriction compliance	User has dietary restrictions	No restricted foods in plan	Dietary filters working correctly – PASS
MP G-003	Meal plan save functionality	Meal plan generated	Plan stored in database	Save functionality working properly – PASS


AI Meal Planner
Generate personalized meal plans based on your goals
View Your Saved Plans
Weight (kg) *
45
Height (cm) *
150
Age *
22
Gender *
Select gender
Dietary Preference *
Select preference
Fitness Goal *
Select goal
Meals Per Day *
3 meals
Calorie Target (optional)
e.g., 2000
Allergies (optional)
bubble tea
Generate Meal Plan
Reset

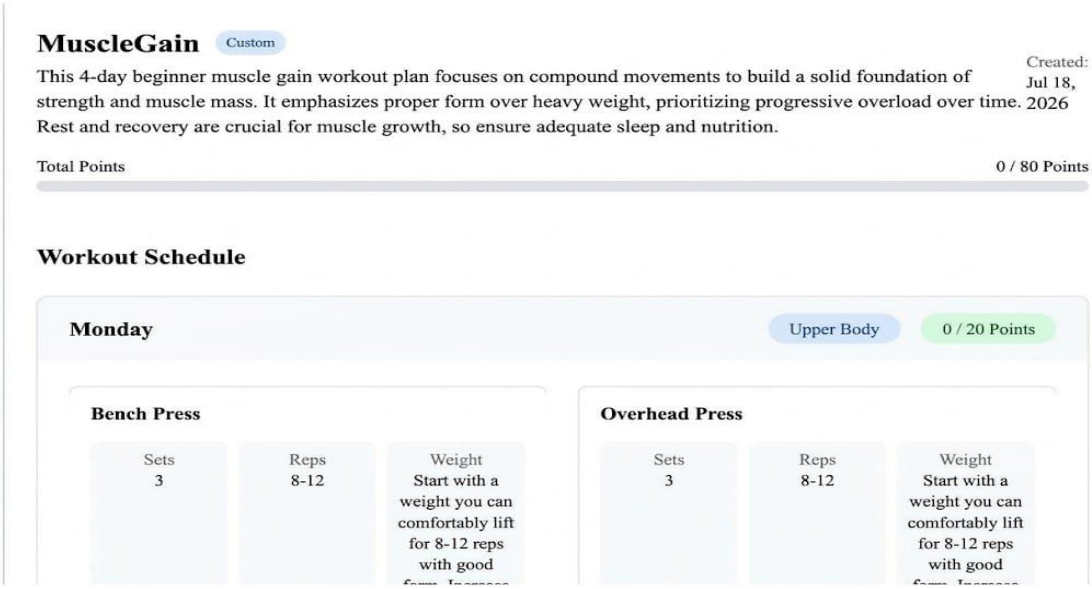

Meal Plan: Omnivore Maintenance Plan
Your Personalized Meal Plan
2026-05-10
Meal plan saved successfully!
A nutrient-dense, balanced 3-meal omnivore plan formulated for a 45kg, 150cm young adult to maintain weight with approximately 1,233 calories. It emphasizes lean proteins, complex carbohydrates, and healthy fats while naturally avoiding bubble tea and processed sugars.
Daily Nutrition Summary
Calories
1233
Protein
85g
Carbs
106g
Fat
53g
Breakfast - Avocado & Egg Toast with Fresh Berries
Eggs - 2 large eggs
Boiled, poached, or scrambled in a non-stick pan using cooking spray.
144 calories
Whole Wheat Bread - 1 slice (approx. 35g)
Toasted until golden brown.
82 calories
Avocado - 35g (approx. 1/4 small avocado)
Lunch - Grilled Chicken Quinoa Bowl
Chicken Breast - 120g (raw weight)
Seasoned with garlic powder, paprika, salt, and grilled or pan-seared.
144 calories
Cooked Quinoa - 130g
Steamed in water or low-sodium vegetable broth.
156 calories
Mixed Roasted Vegetables -
Dinner - Baked Salmon with Sweet Potato & Greens
Atlantic Salmon Fillet - 120g
Baked with herbs, black pepper, and fresh lemon slices.
247 calories
Baked Sweet Potato - 150g
Oven-roasted or microwaved until tender.
130 calories
Steamed Asparagus - 120g
Steamed or blanched until bright green and crisp.

4.2.5 Workout Plan Suggestion Testing

Test ID	Description	Pre-condition	Expected Result	Actual Result
WPS - 001	AI workout plan generation	User profile and goals set	Personalized workout plan created	AI generated appropriate workout plan – PASS
WPS - 002	Fitness level matching	User fitness level defined	Exercises match fitness level	Workout difficulty appropriate – PASS
WPS - 003	Goal-specific workouts	Fitness goals defined	Exercises support user goals	Goal-oriented exercise selection – PASS

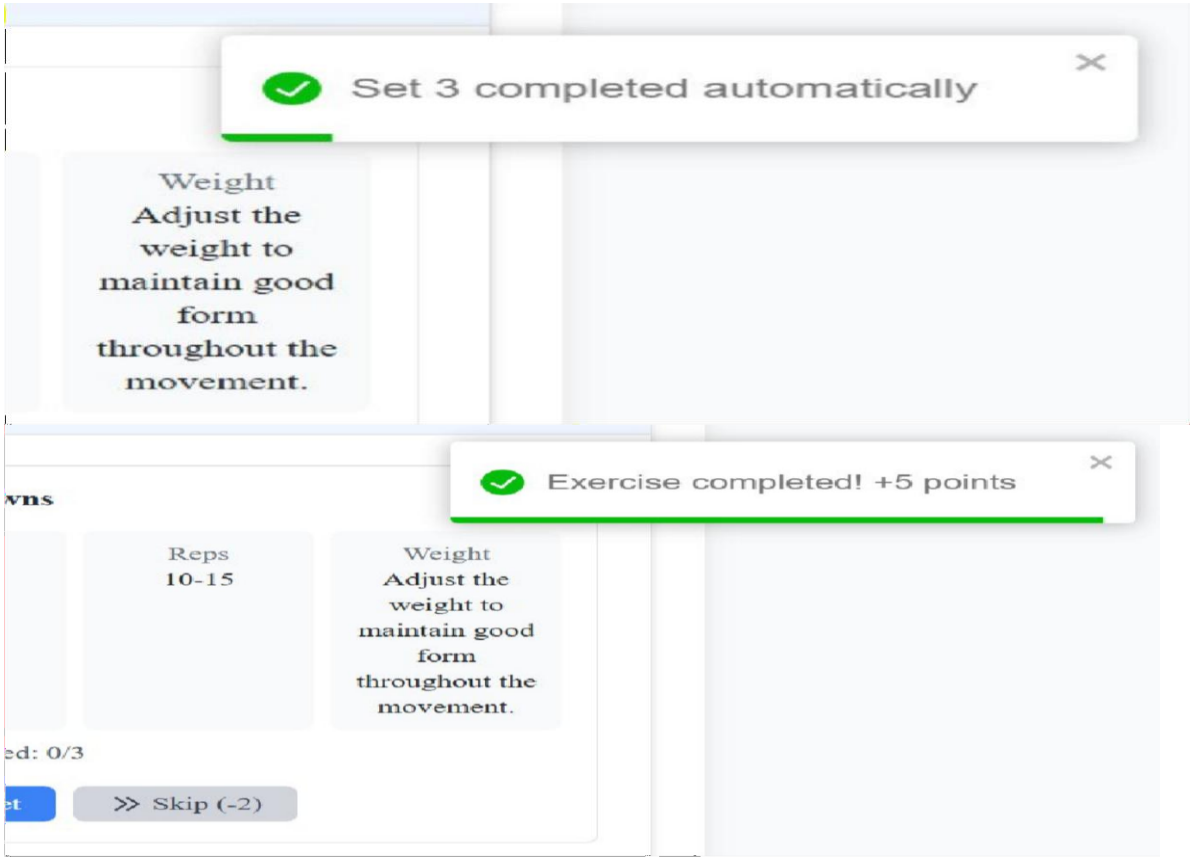



Exercise Plan: 4-Day Muscle Gain Beginner Plan
Your AI-Generated Workout Plan
Save Plan
This 4-day beginner muscle gain workout plan focuses on compound movements to build a solid foundation of strength and muscle mass. It emphasizes proper form over heavy weight, prioritizing progressive overload over time. Rest and recovery are crucial for muscle growth, so ensure adequate sleep and nutrition.
45
Monday
Upper Body
EXERCISE
SETS
REPS
WEIGHT
Bench Press
3
8-12
Start with a weight you can comfortably lift for 8-12 reps with good form. Increase weight gradually each week.


MuscleGain
Custom
This 4-day beginner muscle gain workout plan focuses on compound movements to build a solid foundation of strength and muscle mass. It emphasizes proper form over heavy weight, prioritizing progressive overload over time. Rest and recovery are crucial for muscle growth, so ensure adequate sleep and nutrition.
Created: Jul 18, 2026
Total Points
0 / 80 Points
Workout Schedule
Monday
Upper Body
0 / 20 Points
Bench Press
Sets
3
Reps
8-12
Weight
Start with a weight you can comfortably lift for 8-12 reps with good
Overhead Press
Sets
3
Reps
8-12
Weight
Start with a weight you can comfortably lift for 8-12 reps with good

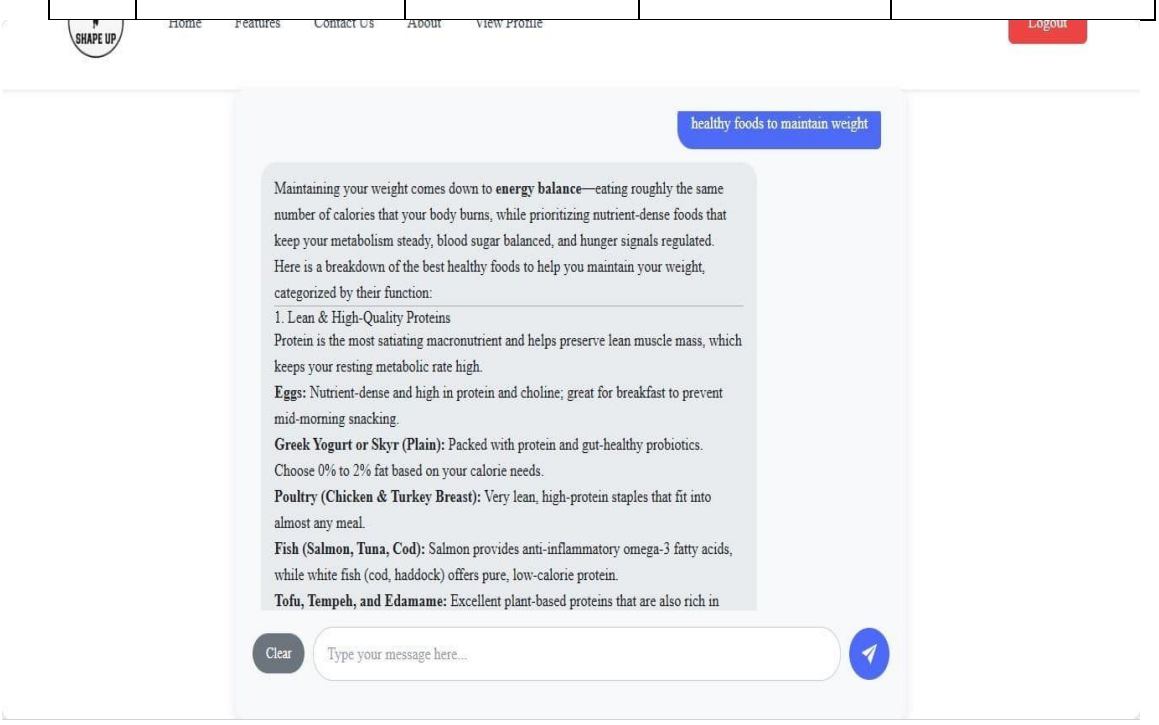
4.2.6 Logging and Progress Monitoring Testing

Test ID	Description	Pre-condition	Expected Result	Actual Result
LPM - 001	Workout completion logging	User completes workout	Activity recorded with timestamp	Workout logging successful - PASS
LPM - 002	Streak tracking	Multiple workout logged	Streak counter updates	Streak system working correctly - PASS



4.2.7 AI Recommendation Logic Testing

Test ID	Description	Pre-condition	Expected Result	Actual Result
AIR -001	Chatbot response accuracy	User asks fitness question	Accurate response provided	Chatbot responses relevant and helpful - PASS
AIR -002	Response time optimization	User requests AI assistance	Response within 5 seconds	Average response time 3.2 seconds – PASS



4.2.8 Cross-Device Responsiveness Testing

Test ID	Description	Pre-condition	Expected Result	Actual Result
CDR - 001	Mobile compatibility	User on smartphone	All features accessible	Mobile interface fully functional - PASS

CDR - 002	Tablet compatibility	User on tablet	Layout optimized for tablet	Tablet layout working properly - PASS
CDR-003	Desktop compatibility	User on desktop	Full functionality available	Desktop version fully functional - PASS
003	compatibility	desktop	available	PASS
CDR - 004	Cross-browser compatibility	Different browsers tested	Consistent functionality	Works across Chrome, Firefox, Safari - PASS

4.2.9 Data Storage and Retrieval Testing

Test ID	Description	Pre-condition	Expected Result	Actual Result
DSR - 001	Data persistence	User creates profile	Data stored permanently	User data persists across sessions – PASS
DSR - 002	Data synchronization	Multiple device access	Data synchronized	Cross-device sync working – PASS
DSR - 003	Data retrieval speed	User requests data	Data loads within 2 seconds	Average load time 1.8 seconds – PASS

4.2.10 UI/UX Usability Testing

Test ID	Description	Pre-condition	Expected Result	Actual Result

UUX-001	Navigation intuitiveness	First-time user access	Features found within 2 minutes	Users navigated easily – PASS
UUX-002	Form completion ease	User fills forms	Forms completed efficiently	Form design user-friendly – PASS
UUX-003	Error message clarity	User encounters errors	Clear actionable messages	Error handling effective – PASS

Complete Testing

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS

5.1. Conclusion

This project was done to create an end-to-end, AI-based fitness and nutrition software that would provide personalized health recommendations to anyone. The visual analytics provides the users with useful information about the fitness process. The smooth integration of other APIs, including the utilization of Google Gemini to access AI assistance to access nutritional information helps users get personalized recommendations based on their own profiles and goals.

The main goals of the project were to develop an integrated platform that consolidates meal planning, exercise suggestion, progress monitoring, and AI support into one easy-to-use application. The goals were to leverage the newest web technology and artificial intelligence to develop a solution that responds to the user's needs without making it too difficult to use.

5.2. Future Recommendations

Although the Fitly AI - Exercise and Meal Planner system successfully fulfills its primary objectives, several enhancements can be integrated in future iterations to increase system performance, user engagement, and health tracking accuracy:

- Wearable Integration: Sync smart watches and fitness bands (Apple Health, Google Fit) to log live biometrics like heart rate, daily steps, and sleep automatically.
- Photo-Based Meal Logging: Capture meal photos to let multimodal AI auto-detect food items, estimate portion sizes, and calculate macros instantly.
- Live Form Correction: Use camera-based pose estimation (Media Pipe) during workouts to deliver real-time feedback on posture and prevent injuries.
- Offline Access: Implement on-device ML models so users can view saved plans and log workouts without an active internet connection.
- Localized Diets & Languages: Support regional cuisines (including South Asian and local Nepali meals) alongside multi-language settings for broader usability.

REFERENCES

[1] Express.js. (n.d.). Express - Fast, unopinionated, minimalist web framework for Node.js. OpenJS Foundation. https://expressjs.com/

[2] Google Cloud. (n.d.). Gemini API documentation: Generative AI for developers. Google. https://ai.google.dev/docs

[3] Jones, M., Bradley, J., & Sakimura, N. (2015). JSON Web Token (JWT) (RFC No. 7519). Internet Engineering Task Force (IETF). https://doi.org/10.17487/RFC7519

[4] MongoDB, Inc. (n.d.). MongoDB Atlas: Fully managed cloud database. MongoDB. https://www.mongodb.com/products/platform/atlas-database

[5] Meta Open Source. (n.d.). React: A JavaScript library for building user interfaces. Meta Platforms. https://react.dev/

[6] Render Services, Inc. (n.d.). Render: Cloud application hosting platform. Render. https://render.com

[7] Tailwind Labs Inc. (n.d.). Tailwind CSS: A utility-first CSS framework for rapid UI development. Tailwind Labs. https://tailwindcss.com/

APPENDICES

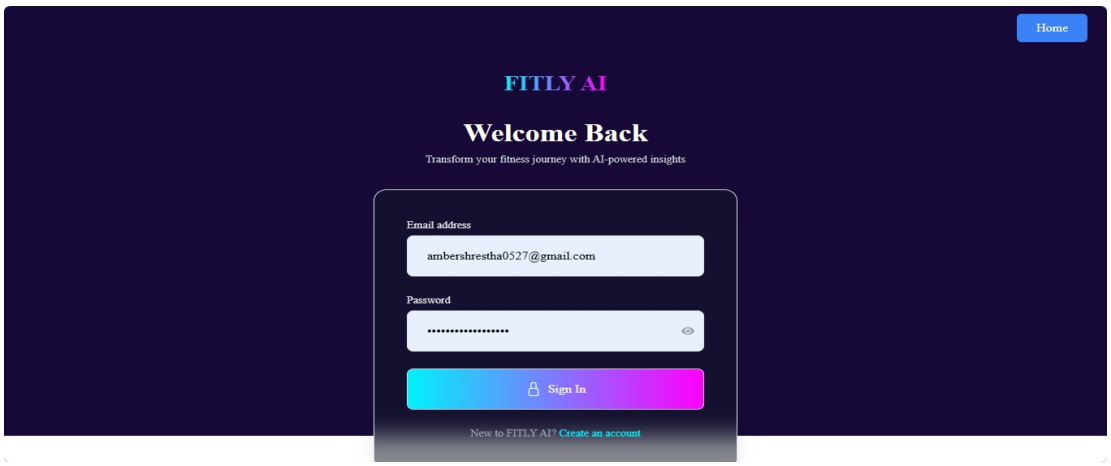


Figure 9 Login Page

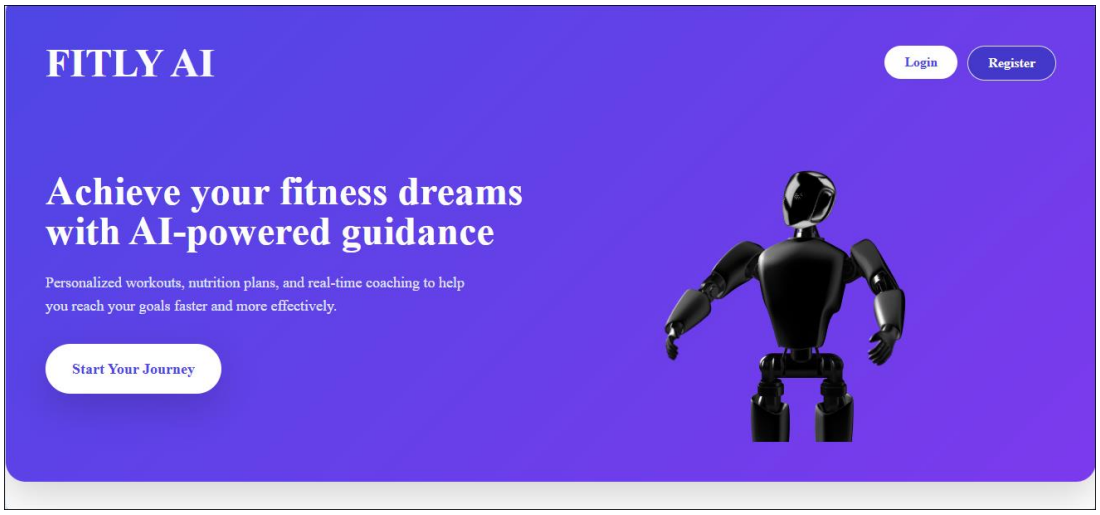


Figure 10 Landing page

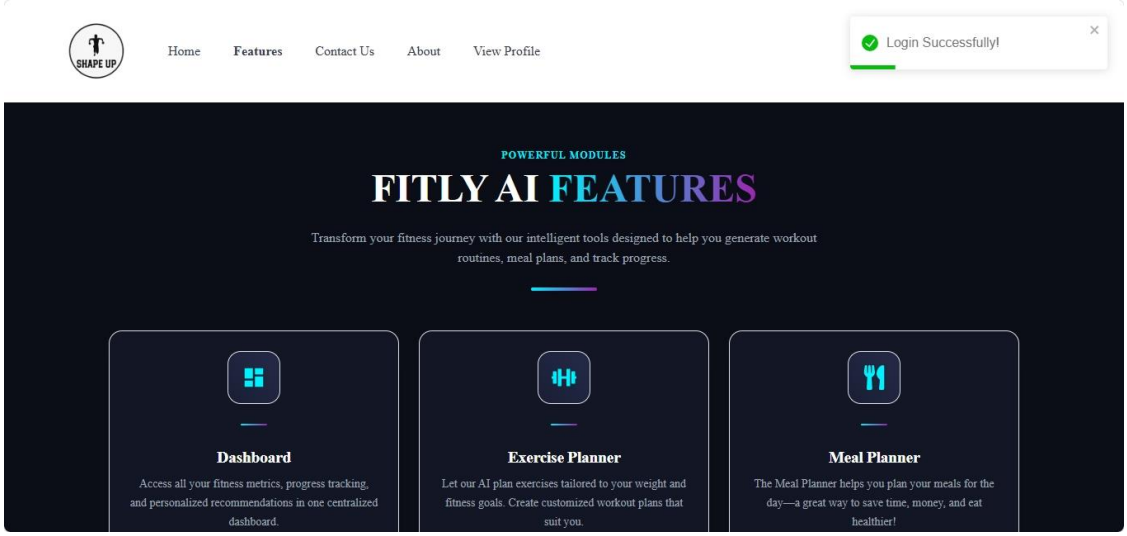


Figure 11 Feature Page

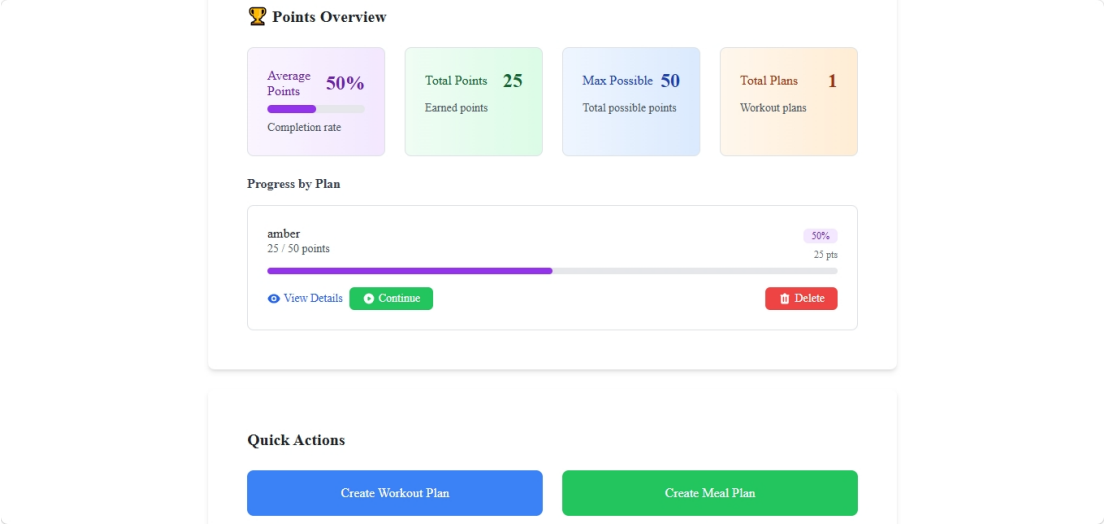


Figure 12 Dashboard Page

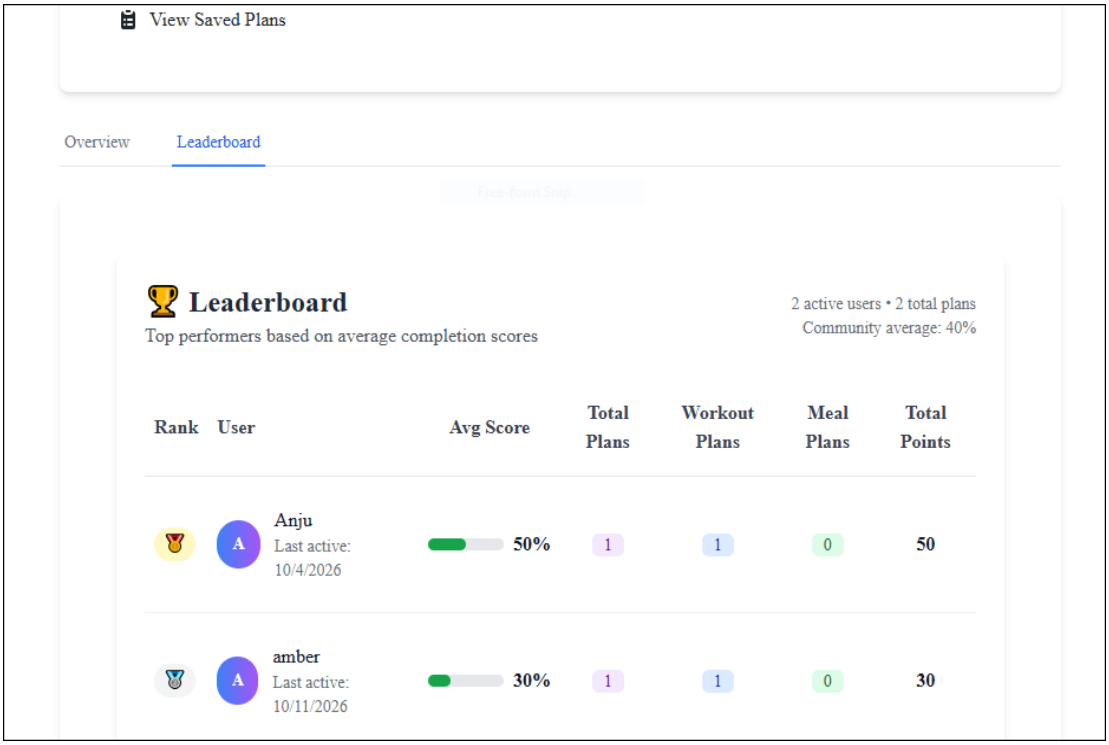


Figure 13 leader Page

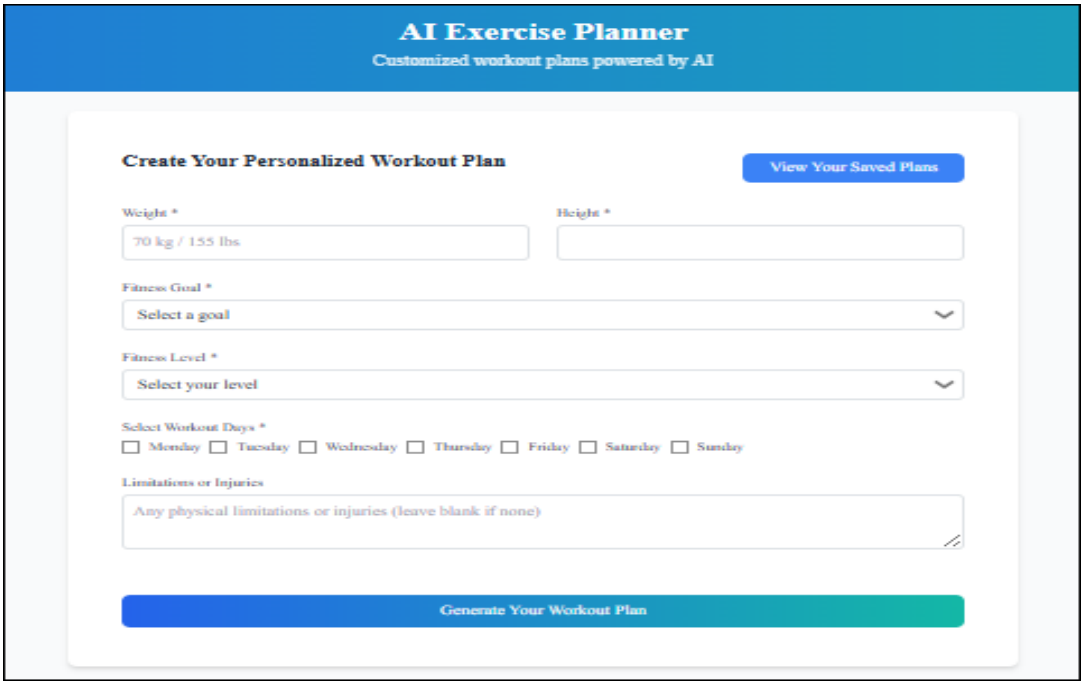


Figure 14 Exercise Planner Page

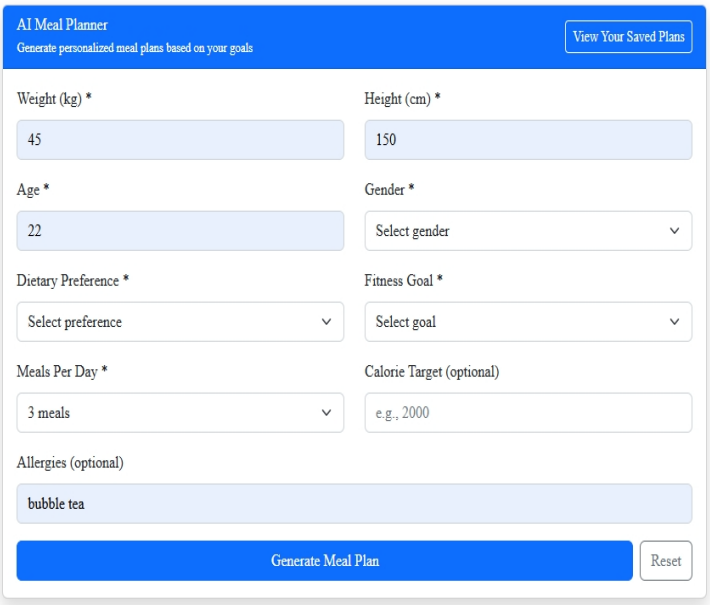


Figure 15 Meal Planner Page

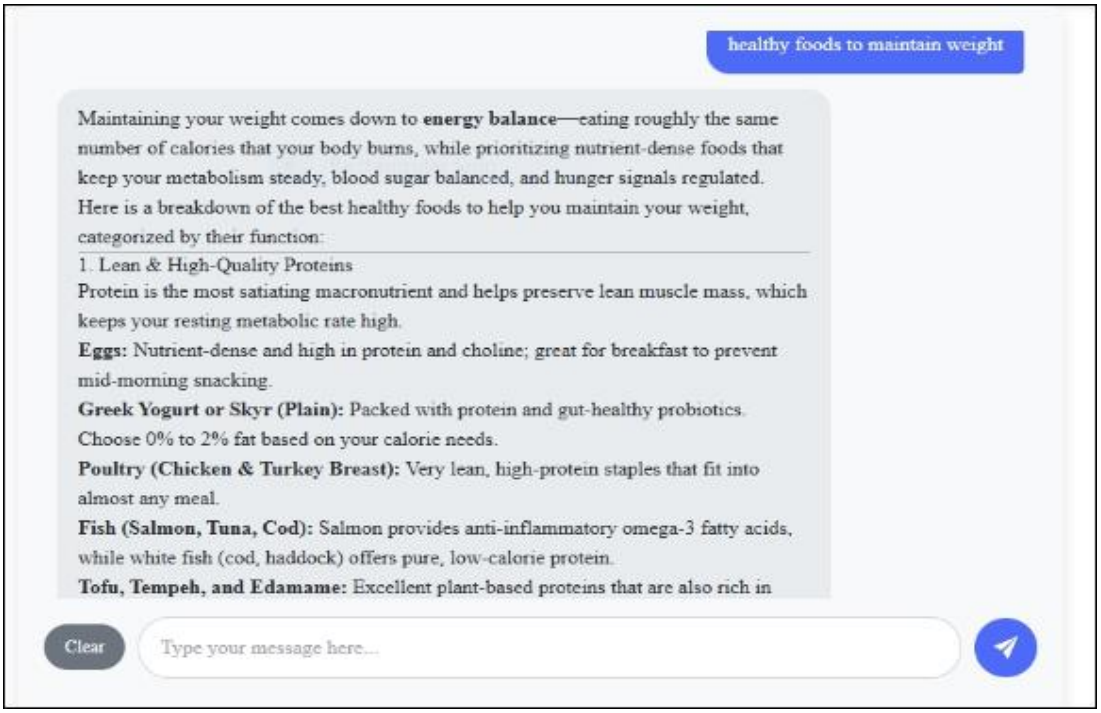


Figure 16 AI Chat-Bot Page